Assignment 4

1. $\begin{pmatrix} 4 & 2 & 0 \\ 4 & 4 & 2 \\ 2 & 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$

Solution:

$$\begin{pmatrix} 4 & 2 & 0 \\ 4 & 4 & 2 \\ 2 & 2 & 3 \end{pmatrix} \xrightarrow{r_2 - r_1} \begin{pmatrix} 4 & 2 & 0 \\ 0 & 2 & 2 \\ 2 & 2 & 3 \end{pmatrix}$$

$$\xrightarrow{r_3 - \frac{1}{2}r_1} \begin{pmatrix} 4 & 2 & 0 \\ 0 & 2 & 2 \\ 0 & 1 & 3 \end{pmatrix}$$

$$\xrightarrow{r_3 - \frac{1}{2}r_2} \begin{pmatrix} 4 & 2 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{pmatrix} = U$$

$$\therefore L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & 1 \end{pmatrix}$$

$\because LUx = b$ i.e. $Ux = c$ $Lc = b$

$\Rightarrow$ ~~$L^{-1}Lc = L^{-1}b$~~

$$\left.\begin{aligned} c_1 &= 2 \\ c_1 + c_2 &= 4 \\ \tfrac{1}{2}c_1 + \tfrac{1}{2}c_2 + c_3 &= 6 \end{aligned}\right\} \Rightarrow \begin{cases} c_1 = 2 \\ c_2 = 2 \\ c_3 = 4 \end{cases}$$

$Ux = c$

$$\Rightarrow \begin{aligned} 4x_1 + 2x_2 &= 2 \\ 2x_2 + 2x_3 &= 2 \\ 2x_3 &= 4 \end{aligned} \Rightarrow \begin{cases} x_1 = 1 \\ x_2 = -1 \\ x_3 = 2 \end{cases}$$

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2. ~~0.0001~~

$$\begin{pmatrix} 0.001 & 2.000 & 3.000 & 1.000 \\ -1.000 & 3.712 & 4.623 & 2.000 \\ -2.000 & 1.072 & 5.643 & 3.000 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} -2.000 & 1.072 & 5.643 & 3.000 \\ -1.000 & 3.712 & 4.623 & 2.000 \\ 0.001 & 2.000 & 3.000 & 1.000 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} -2.000 & 1.072 & 5.643 & 3.000 \\ 0 & 3.176 & 1.815 & 0.500 \\ 0 & 2.000536 & 3.002845 & 1.0015 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} -2.000 & 1.072 & 5.643 & 3.000 \\ 0 & 3.176 & 1.815 & 0.500 \\ 0 & 0 & 1.8680716 & 0.68655416 \end{pmatrix}$$

so $x_3 \approx 0.3675$ $x_2 \approx -0.05102$ $x_1 \approx -0.4904$ (参考)

$\|A\| = 13.266$

$$A^{-1} = \begin{pmatrix} 1.3476 & -0.6801 & -0.1593 \\ -0.3036 & 0.5061 & -0.2532 \\ 0.5353 & -0.3372 & 0.1689 \end{pmatrix} \qquad \|A^{-1}\| = 2.1865$$

$cond(A) = \|A\| \cdot \|A^{-1}\| \approx 26 = 2.6 \times 10^1 \Rightarrow$ only lose 1 digits of accuracy in computing x.

$e = \|x - x^*\|_2$

$= \sqrt{(0.00002)^2}$

$= 0.00002$

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3. $A = A^T$ ~~Gauss~~ Guassian elimination

$$A \rightarrow \begin{pmatrix} a_{11} & a^T \\ 0 & B \end{pmatrix}$$

Prove that B is also a symmetric matrix

Proof: $B = (b_{ij})$ $i, j = 1, 2, \cdots, n-1$

$$b_{ij} = a_{i+1,j+1} - \frac{a_{i+1,1}}{a_{11}} a_{1,j+1}$$

$$b_{ji} = a_{j+1,i+1} - \frac{a_{j+1,1}}{a_{11}} a_{1,i+1}$$

we know $A = A^T \Rightarrow a_{i+1,j+1} = a_{j+1,i+1}$

$$a_{i+1,1} = a_{1,i+1}$$

$$a_{j+1,1} = a_{1,j+1}$$

so $b_{ij} = b_{ji}$

$\Rightarrow B = B^T$

$O(n^3)$ 64倍.

4. ① computer L and U : $\frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n$

② computer $Lc = b$: n^2

③ computer $Ux = c$: n^2 → $O(n^2)$ 16倍.

Totally. $\frac{2}{3}n^3 + 2n^2$

when $n = 2000$, 0.1s.
$k = 1$

Totally: 166.4s

so when $n = 8000$ $k = 100$

$64 \times 0.1s = 6.4s$

$100 \times 0.1s \times 16 = 160s$

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